



ANDROID DEVELOPMENT

Complete Learning Roadmap

Build robust Android apps with Kotlin, modern architecture components, and scalable releases.

Kotlin basics

Jetpack architecture components

State and navigation

Data storage patterns

Testing basics

Compose UI

01

Fundamentals

02

Core Concepts

03

Tools and
Frameworks

04

Real Projects

05

Advanced Topics

5

Learning Stages

20+

Core Topics

10+

Projects

100%

Android



Learning Path Overview

Five progressive stages from fundamentals to production mastery

01

Fundamentals

Kotlin basics

02

Core Concepts

Jetpack architecture components • State and navigation • Data storage patterns • Testing basics

03

Tools and Frameworks

Compose UI • Room database

04

Real Projects

Commerce app module • Realtime chat module • Offline sync module • Analytics instrumentation

05

Advanced Topics

Performance profiling • Battery optimization • Advanced CI/CD for Play Store • Large-team modularization

How to Use This Roadmap

Build and ship first

Complete a small working project in every stage before advancing.

Use official sources

Start with the docs and use secondary resources to reinforce implementation patterns.

Iterate on projects

Refactor earlier work as you learn better architecture, testing, and performance habits.



Stages 01 & 02 - Deep Dive

Fundamentals, core concepts, and architectural foundations

01

Stage 1 - Fundamentals

Estimated: 3-4 weeks | Prerequisites: Basic foundations

Topics

- Kotlin basics
- Activity and Fragment lifecycle
- UI fundamentals
- Gradle fundamentals

Tools & Frameworks

- Kotlin basics

Recommended Projects

- Android Development workshop using Kotlin basics
- Android Development portfolio project with Activity and Fragment lifecycle

Stage 1 Milestone Complete Android Development workshop using Kotlin basics and document the decisions, trade-offs, and validation checks before moving to Stage 2.

02

Stage 2 - Core Concepts

Estimated: 4-5 weeks | Prerequisites: Stage 1 complete

Topics

- Jetpack architecture components
- State and navigation
- Data storage patterns
- Testing basics

Tools & Frameworks

- Jetpack architecture components
- State and navigation
- Data storage patterns
- Testing basics

Recommended Projects

- Android Development workshop using Jetpack architecture components
- Android Development portfolio project with State and navigation

Stage 2 Milestone Complete Android Development workshop using Jetpack architecture components and document the decisions, trade-offs, and validation checks before moving to Stage 3.

Stages 03 & 04 - Deep Dive

Modern tooling, delivery frameworks, and portfolio projects

03

Stage 3 - Tools and Frameworks

Estimated: 4-6 weeks | Prerequisites: Stage 2 complete

Topics

- Compose UI
- Room database
- Dependency injection
- Background work manager

Tools & Frameworks

- Compose UI
- Room database

Recommended Projects

- Android Development workshop using Compose UI
- Android Development portfolio project with Room database

Stage 3 Milestone Complete Android Development workshop using Compose UI and document the decisions, trade-offs, and validation checks before moving to Stage 4.

04

Stage 4 - Real Projects

Estimated: 6-8 weeks | Prerequisites: Stage 3 complete

Topics

- Commerce app module
- Realtime chat module
- Offline sync module
- Analytics instrumentation

Tools & Frameworks

- Commerce app module
- Realtime chat module
- Offline sync module
- Analytics instrumentation

Recommended Projects

- Android Development workshop using Commerce app module
- Android Development portfolio project with Realtime chat module

Stage 4 Milestone Complete Android Development workshop using Commerce app module and document the decisions, trade-offs, and validation checks before moving to Stage 5.

Stage 05 - Advanced Topics & Delivery

Performance, architecture, release automation, and production governance

05

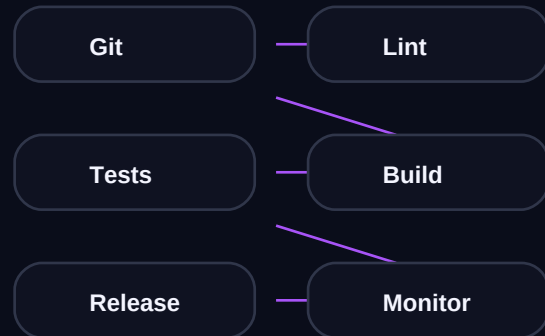
Stage 5 - Advanced Topics

Estimated: Ongoing | Industry-grade operating habits

Advanced Focus

- Performance profiling
- Battery optimization
- Advanced CI/CD for Play Store
- Large-team modularization

CI/CD Pipeline



Stage 5 Milestone

A production-grade android development workflow with measurable performance, quality, and release confidence.

Performance Budget

Set and track concrete engineering thresholds: startup latency, error budget, throughput, cost efficiency, and deployment confidence.



Resources & Practice Questions

Verified links, self-assessment, and community follow-up

FREE

Android Development docs and community guides

Link: <https://developer.android.com/docs>

FREE

Build one portfolio project per stage before moving ahead

Link: <https://roadmap.sh/android>

PAID

Structured specialization track focused on Android Development

Link: <https://roadmap.sh/android>

FREE

Android Development roadmap reference

Link: <https://roadmap.sh/android>

FREE

Official documentation

Link: <https://developer.android.com/docs>

Self-Assessment Practice Questions

- How would you evaluate readiness for Stage 3 in Android Development?
- Which project best proves Stage 4 competency in Android Development?
- What trade-offs appear in Stage 5 decisions for Android Development?